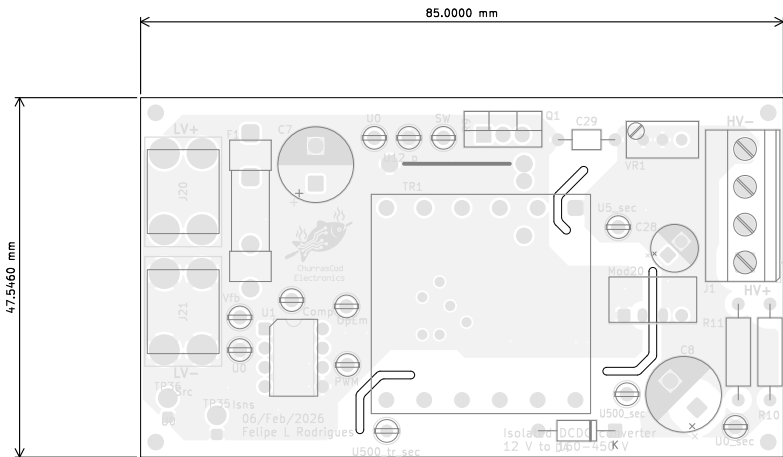


Step-Up Fabrication Document

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

Layer Stack Legend

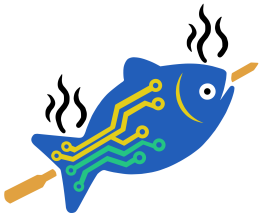
| Material     | Layer                      | Thickness        | Dielectric      | Type         | Gerber |
|--------------|----------------------------|------------------|-----------------|--------------|--------|
| F.Paste      |                            |                  |                 | Paste Mask   |        |
| F.Silkscreen |                            |                  | Direct Printing | Legend       | GBR    |
| F.Mask       |                            | 0.02mm           | Solder Resist   | Solder Mask  | GBR    |
| Copper       | L1 (Ref)                   | 0.07mm (2.00oz)  |                 | Plane        | GBR    |
| Prepreg      |                            | 0.001mm          | FR4_7628        | Dielectric   |        |
| Copper       | Virtual (Top side Jumpers) | 0.035mm (1.00oz) |                 | Copper Layer | GBR    |
| Core         |                            | 1.35mm           | FR4             | Dielectric   |        |
| Copper       | Virtual (Bot side Jumpers) | 0.035mm (1.00oz) |                 | Copper Layer | GBR    |
| Prepreg      |                            | 0.001mm          | FR4             | Dielectric   |        |
| Copper       | L2 (Sig, PWR)              | 0.07mm (2.00oz)  |                 | Signal       | GBR    |
| B.Mask       |                            | 0.02mm           | Solder Resist   | Solder Mask  | GBR    |
| B.Silkscreen |                            |                  | Direct Printing | Legend       | GBR    |
| B.Paste      |                            |                  |                 | Paste Mask   |        |

Total thickness: 1.602mm  
Note: external layer thicknesses are specified after plating

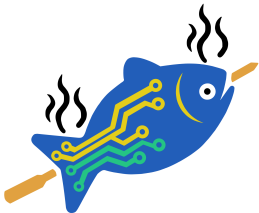
FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

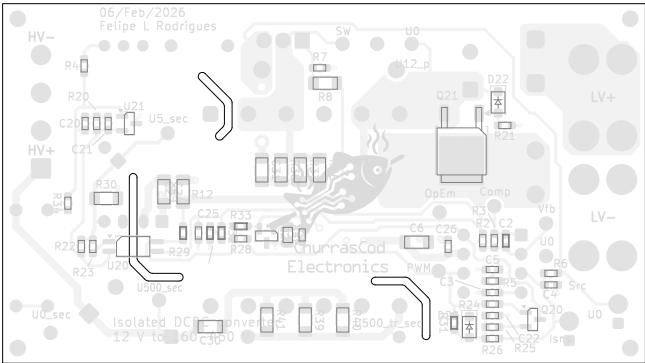
- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts.GBR" SUFFIX.  
  
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.  
  
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:  
  
A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.  
B. Tg 170 C OR EQUIVALENT.  
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NA.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

|                  |                    |
|------------------|--------------------|
| BOARD SIZE       | 85,000 × 47,546 mm |
| BOARD THICKNESS  | 1.602 mm           |
| TRACE WIDTH      | 0.400 mm           |
| TRACE TO TRACE   | -0,000 mm          |
| MIN. HOLE (PTH)  | 0.250 mm           |
| MIN. HOLE (NPTH) | 2.100 mm           |
| ANNULAR RING     | 0.150 mm           |
| COPPER TO HOLE   | 0.200 mm           |
| COPPER TO EDGE   | 0.250 mm           |
| HOLE TO HOLE     | 0.254 mm           |



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|                                                                                      |                                             |                                           |                 |                                            |                                  |
|--------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------|-----------------|--------------------------------------------|----------------------------------|
|  | Comments:                                   | Company:<br>NA                            |                 | Variant:<br>PRELIMINARY                    | Git Hash:<br>421439f             |
|                                                                                      | Sheet Title:<br>Top Fabrication (Scale 1:1) | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                                  |
|                                                                                      | Sheet Path:                                 | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0+ (Unreleased) |
|                                                                                      |                                             |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>1 of 7                 |



All dimensions are in millimeters unless otherwise specified.



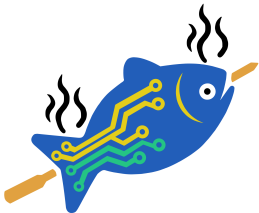
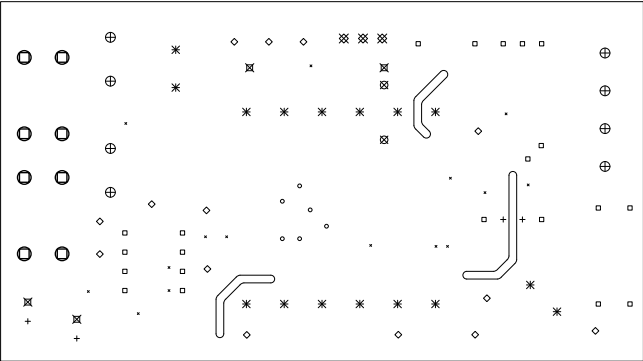
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|--------------------------------------------------------------------------------------|--------------------------------|-----------------------------|-------------|------------|---------------------------|-----------|
|  | Comments:                      |                             | Company:    |            | Variant:                  | Git Hash: |
|                                                                                      |                                |                             | NA          |            | PRELIMINARY               | 421439f   |
|                                                                                      | Sheet Title:                   |                             | Board Name: |            | Project Name:             |           |
|                                                                                      | Bottom Fabrication (Scale 1:1) |                             | Step-Up     |            | Isolated step-up 12:450 V |           |
| Sheet Path:                                                                          |                                | File Name:                  | Designer:   | Date:      | Revision:                 |           |
|                                                                                      |                                | Isolated_12to450V.kicad_pcb | FR          | 2026-02-07 | 2.0.0+ (Unreleased)       |           |
|                                                                                      |                                |                             | Reviewer:   | Size:      | Sheet:                    |           |
|                                                                                      |                                |                             | FR          | A4         | 2 of 7                    |           |

Step-Up Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

| Symbol    | Count | Hole Size          | Plated | Hole Shape | Drill Layer Pair         | Hole Type |
|-----------|-------|--------------------|--------|------------|--------------------------|-----------|
| ×         | 15    | 0,25mm (9,84mils)  | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Via       |
| ○         | 6     | 0,50mm (19,69mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Via       |
| +         | 4     | 0,70mm (27,56mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| □         | 21    | 0,80mm (31,50mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| ◇         | 14    | 0,90mm (35,43mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| ⊠         | 2     | 1,00mm (39,37mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Via       |
| ✱         | 16    | 1,00mm (39,37mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| ⊠         | 4     | 1,10mm (43,31mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| ⊠         | 3     | 1,20mm (47,24mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| ⊕         | 8     | 1,30mm (51,18mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| ○         | 8     | 1,80mm (70,87mils) | PTH    | Round      | L1 (Ref) - L2 (Sig. PWR) | Pad       |
| Total 101 |       |                    |        |            |                          |           |



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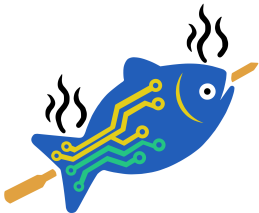
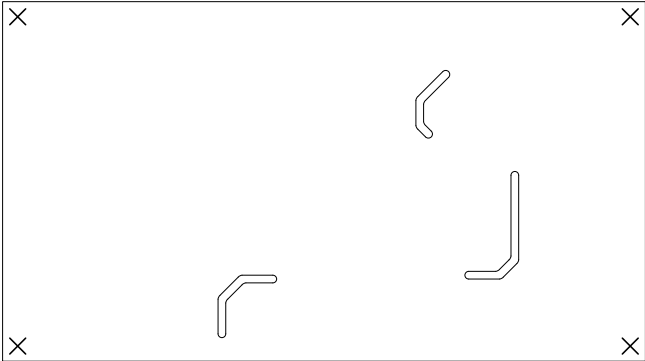
|                                         |                                           |                 |                                            |                                  |
|-----------------------------------------|-------------------------------------------|-----------------|--------------------------------------------|----------------------------------|
| Comments:                               | Company:<br>NA                            |                 | Variant:<br>PRELIMINARY                    | Git Hash:<br>421439f             |
|                                         | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                                  |
| Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0+ (Unreleased) |
| Sheet Path:                             |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>3 of 7                 |

# Step-Up Fabrication Document

## Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

| Symbol | Count   | Hole Size          | Plated | Hole Shape | Drill Layer Pair         | Hole Type  |
|--------|---------|--------------------|--------|------------|--------------------------|------------|
| X      | 4       | 2.10mm (82.68mils) | NPTH   | Round      | L1 (Ref) - L2 (Sig, PWR) | Mechanical |
|        | Total 4 |                    |        |            |                          |            |



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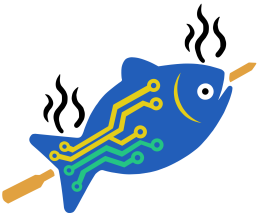
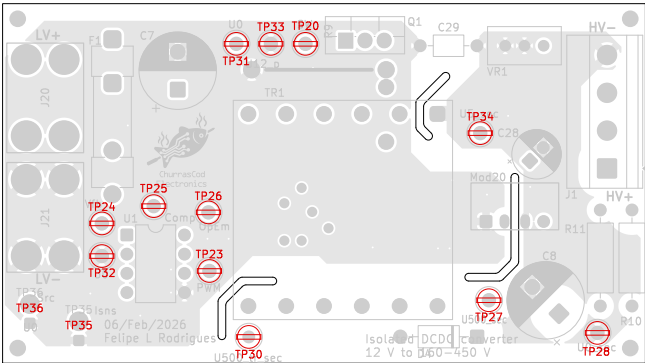
|                                         |                                           |                 |                                            |                                  |  |
|-----------------------------------------|-------------------------------------------|-----------------|--------------------------------------------|----------------------------------|--|
| Comments:                               | Company:<br>NA                            |                 | Variant:<br>PRELIMINARY                    | Git Hash:<br>421439f             |  |
|                                         | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                                  |  |
| Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0+ (Unreleased) |  |
| Sheet Path:                             |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>4 of 7                 |  |

# Step-Up Fabrication Document

## Top Test Points (Scale 1:1)

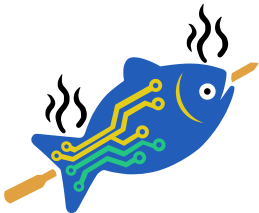
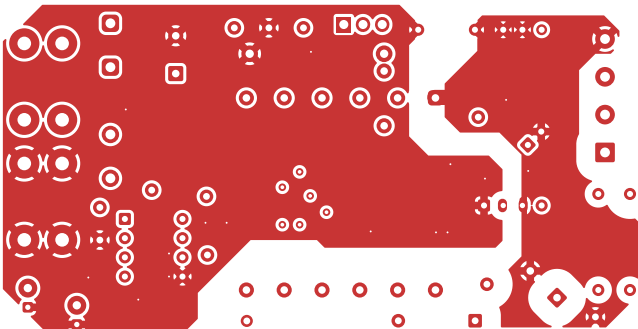
| Ref. | Net         | X [mm] | Y [mm] |
|------|-------------|--------|--------|
| TP20 | SW          | 42.05  | 44.66  |
| TP23 | PWM         | 29.35  | 14.62  |
| TP24 | Vfb         | 15.13  | 20.91  |
| TP25 | Comp        | 21.98  | 23.20  |
| TP26 | OpEm        | 29.22  | 22.37  |
| TP27 | U500_sec    | 66.31  | 10.75  |
| TP28 | U0_sec      | 80.66  | 6.43   |
| TP30 | U500_tr_sec | 34.56  | 5.90   |
| TP31 | U12_p       | 32.91  | 44.66  |
| TP32 | U0          | 15.13  | 16.59  |
| TP33 | U0          | 37.48  | 44.66  |
| TP34 | U5_sec      | 65.16  | 32.85  |
| TP35 | Isns,U0     | 12.08  | 6.70   |
| TP36 | Src,U0      | 5.60   | 8.97   |

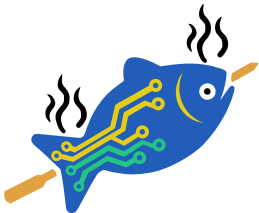
All dimensions are in millimeters unless otherwise specified.



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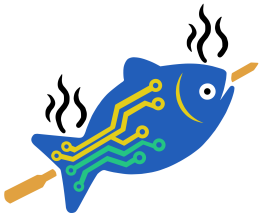
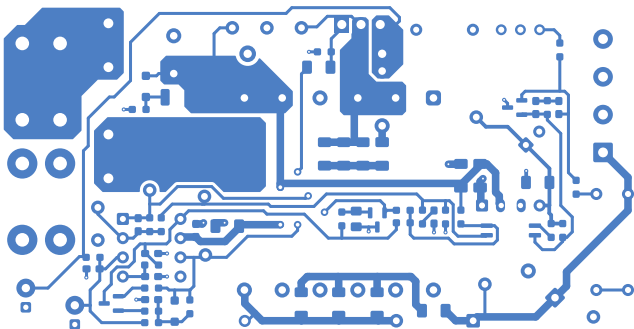
|             |                                             |                                           |                                            |                     |                                  |
|-------------|---------------------------------------------|-------------------------------------------|--------------------------------------------|---------------------|----------------------------------|
| Comments:   | Company:<br>NA                              |                                           | Variant:<br>PRELIMINARY                    |                     | Git Hash:<br>421439f             |
|             | Board Name:<br>Step-Up                      |                                           | Project Name:<br>Isolated step-up 12:450 V |                     |                                  |
|             | Sheet Title:<br>Top Test Points (Scale 1:1) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR                            | Date:<br>2026-02-07 | Revision:<br>2.0.0+ (Unreleased) |
| Sheet Path: |                                             | Reviewer:<br>FR                           | Size:<br>A4                                | Sheet:<br>5 of 7    |                                  |



|                                                                                      |                                  |  |                                           |                 |                                            |                                  |
|--------------------------------------------------------------------------------------|----------------------------------|--|-------------------------------------------|-----------------|--------------------------------------------|----------------------------------|
|  | Comments:                        |  | Company:<br>NA                            |                 | Variant:<br>PRELIMINARY                    | Git Hash:<br>421439f             |
|                                                                                      | Sheet Title:<br>F.Cu (Scale 1:1) |  | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                                  |
|                                                                                      | Sheet Path:                      |  | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0+ (Unreleased) |
|                                                                                      |                                  |  | Reviewer:<br>FR                           | Size:<br>A4     | Sheet:<br>6 of 7                           |                                  |

Step-Up Fabrication Document

B.Cu (Scale 1:1)



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|                                  |                                           |                 |                                            |                                  |
|----------------------------------|-------------------------------------------|-----------------|--------------------------------------------|----------------------------------|
| Comments:                        | Company:<br>NA                            |                 | Variant:<br>PRELIMINARY                    | Git Hash:<br>421439f             |
|                                  | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                                  |
| Sheet Title:<br>B.Cu (Scale 1:1) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0+ (Unreleased) |
| Sheet Path:                      |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>7 of 7                 |